Non Graded Lab Assignment 2 – CSD 207, OOP in Java

For this assignment, you will solve problems based on what you have learned in Chapters: Elementary Programming, and Selections.

Instructions

- Review the chapters from the book and class notes.
- There are 5 questions in this assignment.
- Assignment submitted after due date will not be evaluated.

Due Date and Time: August 10, 2018, in class.

Submitting this Assignment

You do not have to submit this assignment. When you have completed solving all questions, show them to the Instructor.

Grading Criteria

There are no marks to this assignment.

Note: Use class JOptionPane to take input/output wherever required.

Questions:

1. **Write the program which lets the user to enter the interest rate, number of years, and loan amount and computes monthly payment. And, display the monthly payment with the accuracy of two decimal places.**

$$monthlyPayment = \frac{loanAmount \times monthlyInterestRate}{1 - \frac{1}{(1 + monthlyInterestRate)^{numberOfYears \times 12}}}$$

2. **Write a program that randomly generates a lottery of a two-digit number, prompts the user to enter a two-digit number, and determines whether the user wins according to the following rule:**
   - **If the user input matches the lottery in exact order, the award is $10,000.**
   - **If the user input matches the lottery, the award is $3,000.**

- **If one digit in the user input matches a digit in the lottery, the award is $1,000.**

3. **Write a program that displays current time in GMT in the format hour:minute:second such as 1:45:19.**
   The currentTimeMillis method in the System class returns the current time in milliseconds since the midnight, January 1, 1970 GMT. (1970 was the year when the Unix operating system was formally introduced.) You can use this method to obtain the current time, and then compute the current second, minute, and hour as follows.
   (use only arithematic operators to calculate the time)

4. **Body Mass Index (BMI) is a measure of health on weight. It can be calculated by taking your weight in kilograms and dividing by the square of your height in meters. The interpretation of BMI for people 16 years or older is as follows:**

BMI	Interpretation
BMI < 18.5	Underweight
18.5 <= BMI < 25.0	Normal
25.0 <= BMI < 30.0	Overweight
30.0 <= BMI	Obese

5. **Write a program that prompts the user to enter the filing status and taxable income and computes the tax.**

- **The US federal personal income tax is calculated based on the filing status and taxable income. There are four filing statuses: single filers, married filing jointly, married filing separately, and head of household. The tax rates are shown in Table.**

Marginal Tax Rate	*Single*	*Married Filing Jointly or Qualifying Widow(er)*	*Married Filing Separately*	*Head of Household*
10%	$0 – $8,350	$0 – $16,700	$0 – $8,350	$0 – $11,950
15%	$8,351 – $33,950	$16,701 – $67,900	$8,351 – $33,950	$11,951 – $45,500
25%	$33,951 – $82,250	$67,901 – $137,050	$33,951 – $68,525	$45,501 – $117,450
28%	$82,251 – $171,550	$137,051 – $208,850	$68,526 – $104,425	$117,451 – $190,200
33%	$171,551 – $372,950	$208,851 – $372,950	$104,426 – $186,475	$190,201 – $372,950
35%	$372,951+	$372,951+	$186,476+	$372,951+